

Research Note 02

Calendar-Aware Forecasting: EWMA Baseline and an Explicit Holiday Model

Traffic Forecasting Project

June 14, 2026

Abstract

Building on Note 01 (the 4-week seasonal baseline explains $\sim 94\%$ of hourly flow variance), we push past the $R^2 \approx 0.94$ wall with two interpretable, leakage-safe refinements. (1) An **EWMA weekly baseline** replaces the flat 4-week mean with a single learned decay, capturing the “recent weeks matter more” effect. (2) An **explicit holiday model** – a pooled per-(holiday, hour-of-day) multiplicative factor on the baseline – targets the fact that holidays are 8% of hours but $\sim 23\%$ of squared error. In a rolling-origin backtest on 2019, the combined model reaches $R^2 = \mathbf{0.9493}$ (vs. 0.9404 flat), cutting holiday-window RMSE from 67 to 48 veh/h while leaving ordinary hours untouched. Per-sensor holiday factors reveal real corridor heterogeneity (recreational routes *gain* traffic) but add little with only two training years. We then bound the headroom of every endogenous lever – autoregressive (ARIMA-style) residual modeling and learned calendar/sensor features (ridge/XGBoost) – and find all of it small ($\leq +0.003 R^2$). An oracle floor analysis shows our out-of-sample model already matches the best *in-sample* seasonal+holiday fit, so the residual is dominated by irreducible noise: the one-week-ahead problem is, for this slice, **effectively solved** ($R^2 = 0.949$, weekly-aggregate MAPE 3.55%). The open questions are scope – a small weather lever, and generalization (2020, post-2022, other regions, speed).

Key findings

- **EWMA baseline** (one decay parameter $\alpha \approx 0.8$): R^2 $0.9404 \rightarrow 0.9435$. Free weekly weights do no better – one knob suffices.
- A YoY-growth-adjusted “last-year” term is **secretly a holiday model**: its benefit concentrates almost entirely on holiday windows.
- **Explicit holiday model**: holiday-window RMSE $67 \rightarrow 48$ veh/h (-29%), *surgical* (ordinary hours unchanged), and **stacks** with EWMA $\Rightarrow R^2 = 0.9493$.
- Holiday response is **heterogeneous** across corridors, but with two training years the per-sensor refinement is marginal ($+0.0003 R^2$) – a “needs more data” win.
- **Autoregressive headroom is small**. Recent-lag predictability is strong at $h=1$ ($R^2=0.63$) but collapses by a week out (aggregate ceiling $+0.0027$); weekly aggregation is cleaner (ACF 0.30) but caps at $+0.0010$.

- **ML on residual features does not help:** sub-daily calendar features are zero by construction; week-of-year is tiny and redundant; high-cardinality features overfit. The escalation to XGBoost is unwarranted.
- **We are at the practical ceiling.** The out-of-sample model ($R^2=0.949$) already exceeds the best *in-sample* static seasonal+ holiday oracle; the residual is dominated by irreducible noise. Weekly-aggregate MAPE is 3.55% (hourly MAPE is misleading – a low-flow denominator artifact).

1 Setup and the week-ahead constraint (recap)

We forecast hourly freeway *flow* one week ahead ($H = 168$) for $N = 2,352$ Greater Bay Area sensors, pre-COVID 2017–2019 (see Note 01 for data details). The seasonal baseline is $b_{i,t} = \frac{1}{4} \sum_{k=1}^4 y_{i,t-Pk}$, $P = 168$. Note 01 established the key constraint: for a one-week-ahead target $s = t + h$ at origin t , only *weekly* lags ($s - Pk \leq t$) are available across the whole horizon, so every model here uses weekly-aligned information only. All results below are a **rolling-origin backtest on 2019** (359 origins, stride 24h), with any learned parameters fit only on 2017–2018.

2 An EWMA weekly baseline

Note 01 showed that a seasonal-residual AR was, in effect, re-weighting the flat baseline toward recent weeks. We make that explicit with an EWMA baseline

$$b_{i,t}^{\text{ewma}} = \sum_{k=1}^L w_k y_{i,t-Pk}, \quad w_k \propto \alpha^{k-1}, \quad \sum_k w_k = 1. \quad (1)$$

Selecting α on the training years gives $\alpha \approx 0.8$ ($L = 8$), with weights decaying 0.24, 0.19, $\dots$, 0.05. This lifts R^2 from 0.9404 to 0.9435 (Table 1). Fitting eight *free* weekly weights by least squares does no better – a single decay parameter captures all the available linear weekly-lag signal.

3 From a “year-ago” term to an explicit holiday model

We tested a year-over-year idea: scale the value one year ago by the YoY growth ratio, $g_{i,t} y_{i,t-52P}$ with $g = (\text{recent level})/(\text{level a year ago})$, and blend with the baseline. Two lessons emerged:

- A hardcoded 50/50 blend *hurts* (−4% RMSE): the single year-ago value is a high-variance predictor (standalone RMSE 56 vs. 40), so over-weighting it injects noise. At the *optimal* weight (≈ 0.28) it helps by +2.6%.
- That help is **concentrated on holidays**. Bucketing the 2019 error: the baseline is fine on ordinary hours (RMSE 37) but terrible on holiday windows (RMSE 67); the year-ago term improves holidays by +8% and ordinary hours by only +1%. The year-ago value is implicitly a (blurry) holiday signal.

This is decisive: **holiday windows are $\sim 8\%$ of hours but $\sim 23\%$ of total squared error**. They are also perfectly known in advance. So we model them *explicitly* instead of via a noisy proxy.

3.1 The model

For each special day – a federal holiday identity together with its $\{-1, 0, +1\}$ travel-day offset – and each hour-of-day, we learn a single pooled multiplicative factor

$$\mathbf{factor}[\text{key}, \text{hod}] = \frac{\sum_{i, t \in \text{train}} y_{i,t}}{\sum_{i, t \in \text{train}} b_{i,t}}, \quad \hat{y}_{i,s} = b_{i,s} \cdot \mathbf{factor}[\text{key}(s), \text{hod}(s)] \quad (2)$$

on special days (and $\hat{y} = b$ otherwise). Pooling over all 2,352 sensors denoises the factor; keying by holiday *identity* (not a fixed 52-week lag) aligns holidays correctly; the per-hour-of-day resolution captures the daily re-shaping. Figure 1 shows the learned factors: holidays crush the morning commute to $\sim 0.35\times$ and the evening to $\sim 0.7\times$, with a late-night *bump* above 1 (holiday-eve travel).

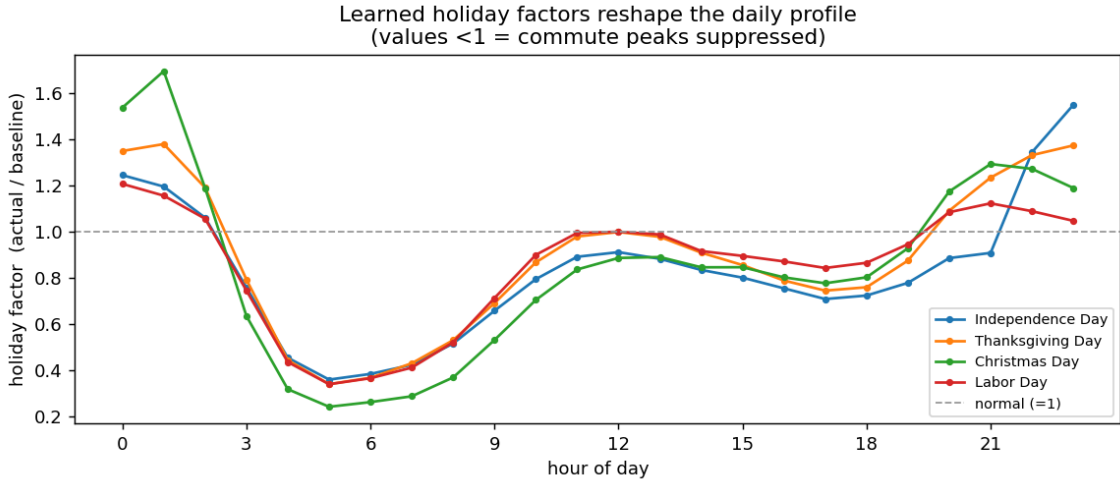


Figure 1: Learned pooled holiday factors by hour-of-day. Values < 1 suppress the commute peaks; the late-evening values > 1 reflect holiday-eve travel.

3.2 Results

Table 1 gives the full progression. The holiday model cuts holiday-window RMSE from 67 to 48 veh/h (-29%), leaves ordinary hours exactly unchanged (it only touches special days), and **stacks** with the EWMA baseline because the two fixes are orthogonal – EWMA improves ordinary hours via recency weighting, the holiday model fixes holidays. Combined: $R^2 = 0.9493$, our first clear step past the baseline wall.

Table 1: Rolling-origin backtest, 2019 (359 origins $\times H=168$, holiday factors from 2017–18). RMSE in veh/h.

Model	R^2 (all)	RMSE all	RMSE ordinary	RMSE holiday
Flat baseline	0.9404	40.14	37.06	66.96
EWMA $\alpha=0.8$	0.9435	39.10	35.98	65.99
Flat + holiday	0.9460	38.20	37.06	49.88
EWMA + holiday (pooled)	0.9493	37.01	35.98	47.71
EWMA + holiday (per-sensor)	0.9496	36.91	35.98	46.69

3.3 Corridor heterogeneity

A pooled factor assumes every corridor reacts to a holiday identically. It does not: Figure 2 shows the distribution of per-sensor holiday *scale* (a per-sensor multiplier on the pooled factor, shrunk toward 1). On Thanksgiving, 31% of corridors *gain* traffic (scale up to $\times 1.5$ on recreational routes such as US-101 N), while commute routes drop. Adding per-sensor scales (shrinkage $\tau = 200$) improves R^2 only from 0.9493 to 0.9496: with just two holiday instances per sensor the estimates are noisy and must be shrunk heavily toward the pooled value. This is a “needs more years” **refinement** – it should pay off once post-2022 data is added.

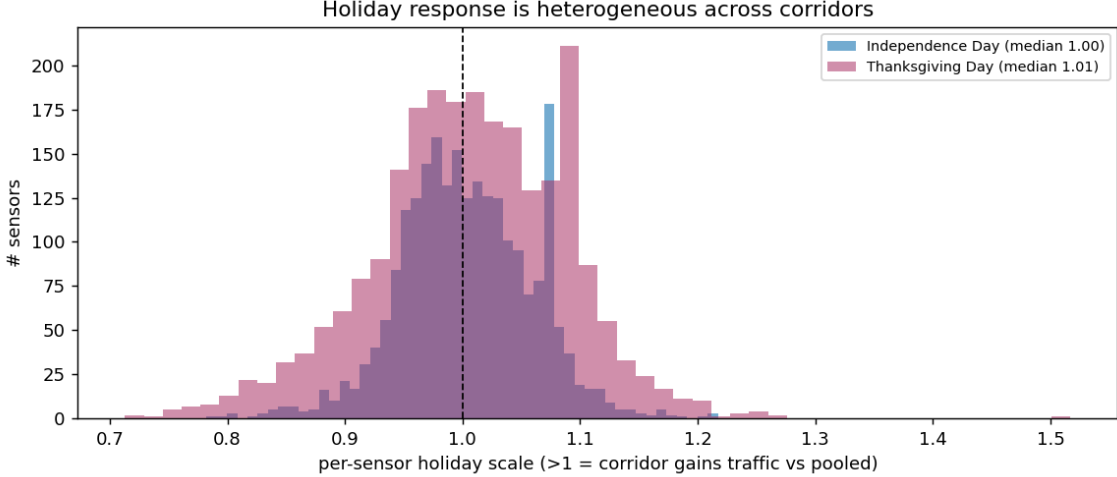


Figure 2: Per-sensor holiday scale (multiplier on the pooled factor). The spread, especially Thanksgiving’s gain tail > 1 , is real corridor heterogeneity, but noisy at two training instances.

4 Does autoregression help? Bounding the residual’s own predictability

Every gain so far comes from the deterministic level (baseline, EWMA, holiday); none exploits the residual’s *own* dynamics, despite its strong short-range autocorrelation (Note 01: lag-1 ≈ 0.81). Does an autoregressive / ARIMA correction on the residual add anything under our **week-aggregate** error metric? We bound it two ways; both are small.

4.1 Recent (sub-weekly) lags help only at short horizons

At origin t forecasting $s = t + h$, a lag- ℓ residual $r_{i,s-\ell}$ is observed iff $\ell \geq h$: a one-day lag is usable only for $h \leq 24$, a one-hour lag only for $h = 1$. The freshest usable information ages with the horizon (and a recursive AR(1) is useless, decaying as $0.81^h \rightarrow 0$). The natural model is a *direct, per-horizon* regression $\hat{r}_{i,t+h} = f_h(\{r_{i,t+h-\ell} : \ell \geq h\})$ added to the baseline.

We measured its ceiling: for each h , the residual variance explained by the best linear predictor on the available lags (train 2017–18, test 2019; Figure 3). It is strong but *collapses* with horizon – $R^2 = 0.63$ at $h=1$, 0.09 at $h=24$, 0.009 at $h=168$ – averaging 0.15 over the first day but only 0.02 beyond it. Scaled by the residual’s 6.6% share of total variance and averaged over the week, the best-case lift in **aggregate** R^2 is just +**0.0027** ($0.9493 \rightarrow \sim 0.9520$) – and that is an unreachable

ceiling. The signal is front-loaded into the first day, so a week-aggregate metric dilutes it roughly 7:1. Worth it only if short horizons (day-ahead, nowcasting) are scored explicitly.

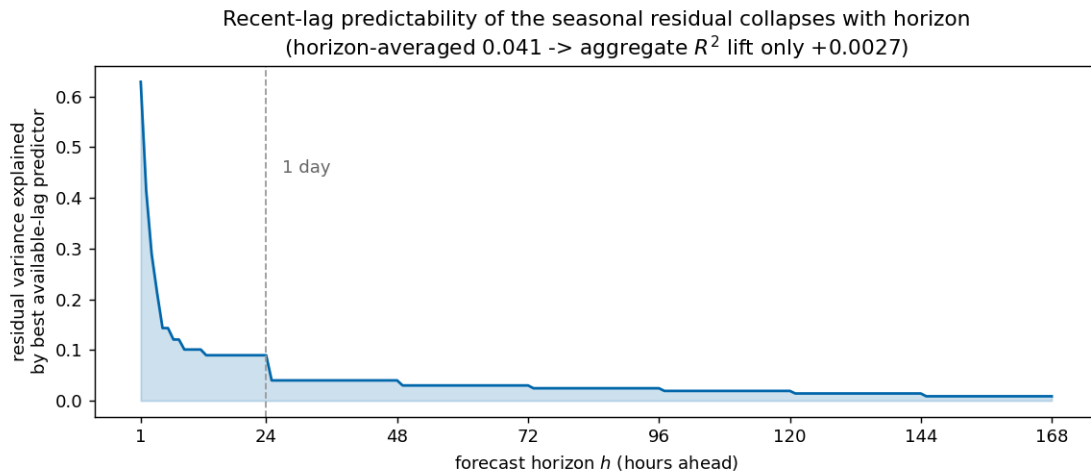


Figure 3: Residual variance recoverable from available lags vs. forecast horizon. Huge at $h=1$ (0.63), gone by a week out – the value lives in the first day and is diluted by a week-aggregate metric.

4.2 Weekly aggregation: cleaner signal, still small

A week-*level* forecast would instead help all horizons uniformly, escaping the dilution. We aggregate the hourly residual to a weekly mean $m_{i,w}$ (within-week noise cancels), fit an AR on the weekly series, and would re-distribute it by historical hourly shape. Aggregation does sharpen the signal: the weekly- aggregate ACF at one week is **+0.295** (vs. 0.094 at the hourly lag-168), with a genuine +0.36 load on last week. But only **14.8%** of the residual variance is week-level coherent – the rest is within-week noise that cancels and therefore cannot be forecast – and the weekly series is only $\sim 10\%$ predictable out-of-sample. The best-case aggregate lift is **+0.0010**, much of it already captured by the EWMA baseline’s recency weighting.

4.3 Verdict and next directions

Bounding the autoregressive headroom from both ends – sub-weekly (+0.0027, diluted) and weekly (+0.0010, redundant) – gives the same answer: **small**. The cause is structural: beyond the seasonal baseline and holidays, the residual is $\sim 90\%$ within-week idiosyncratic noise (incidents, weather, one-off events). Traffic is very regular week to week, so its own past carries little extra; we consider the ARIMA avenue **closed for the week- aggregate metric**. The remaining leverage is **exogenous, all-horizon signals** that inject new information:

- **Weather** (top): precipitation/temperature, forecastable a week ahead, depresses flow across a whole week – the all-horizon shape that made holidays pay.
- Special events, construction and closures (calendar-known); long-run growth/trend.
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5 Do learned residual features (ridge / XGBoost) help?

Before concluding, we test the natural ML escalation: a model (ridge, or XGBoost for interactions) on the seasonal residual with calendar and sensor-type fixed effects – hour-of-week, day-of-week, hour-of-day, week-of-year, and lane count. For a categorical feature the best predictor is its group mean, so the out-of-sample variance explained by group means (fit 2017–18, test 2019) *upper-bounds* what any such model can achieve. The result (Table 2) is decisive: the sub-daily calendar features explain ≈ 0 **by construction** – the baseline already *is* the per-sensor hour-of-week mean, so the residual is orthogonal to them. Only week-of-year carries any signal (+0.0005 aggregate), it is largely redundant with the holiday model, and its sensor-type interaction (XGBoost’s supposed edge) adds nothing; the high-cardinality week-of-year $\times$ sensor feature *overfits* to -0.045 out-of-sample. ML cannot manufacture signal that the construction already removed.

Table 2: Out-of-sample residual variance explained by each feature’s group means (upper bound for any ridge/XGBoost model on that feature). Test 2019.

Feature on the residual	OOS residual R^2	Aggregate R^2 lift
hour-of-week / day-of-week / hour-of-day	≤ 0	0
month-of-year (12)	+0.001	+0.0001
week-of-year (52)	+0.008	+0.0005
week-of-year $\times$ lanes	+0.009	+0.0005
week-of-year $\times$ sensor	-0.045	(overfits)

6 How close are we to the ceiling? Floor and MAPE

Two final checks ask whether the problem is, for practical purposes, solved.

6.1 An irreducible-noise floor

We estimate the variance no week-ahead model can explain by fitting *oracle* structure *in-sample* on the 2019 test year itself (perfect parameters, no estimation error – an upper bound on achievable skill). Table 3 gives the result. A perfect static weekly-seasonal mean reaches only $R^2 = 0.938$, and adding an oracle holiday effect 0.943 – yet our *out-of-sample* EWMA+holiday model already scores 0.949, **above the in-sample static oracle**. The reason is informative: the trailing EWMA tracks *within-year* nonstationarity (sub-annual level drift) that a single annual mean cannot. Together with §6–§7 (no autoregressive or calendar/sensor feature recovers more than $\sim +0.003$), this is strong evidence that the remaining $\sim 5\%$ of variance is **dominated by irreducible idiosyncratic fluctuation** – incidents, weather, one-off events – not learnable structure. We are near the practical ceiling for *endogenous* information; the one untested exogenous lever is weather (Note 03 scope), expected to be small for *flow*.

Table 3: Oracle in-sample fits on 2019 (upper bounds), vs. our out-of-sample model.

Predictor (oracle = fit on test year)	R^2
Oracle static weekly-seasonal mean	0.9378
+ oracle holiday effect	0.9433
Our out-of-sample EWMA + holiday	0.9493

6.2 The metric: weekly-aggregate MAPE

The project’s headline metric is the weekly-aggregate MAPE (per sensor-week total). Under it, EWMA+holiday scores **3.55%** (vs. 3.68% flat) – a strong number. By contrast the *hourly* MAPE looks alarming (82%), but Figure 4 shows this is entirely a small-denominator artifact: low-flow night hours (< 50 veh/h, 13% of points) carry a 583% MAPE, while peak hours (> 300 veh/h, the volume that matters) sit at 6.7%. This both confirms the model is accurate where volume is meaningful and *vindicates aggregating to the week*, which removes the pathology. (A volume-weighted or sMAPE metric would do the same at the hourly level.)

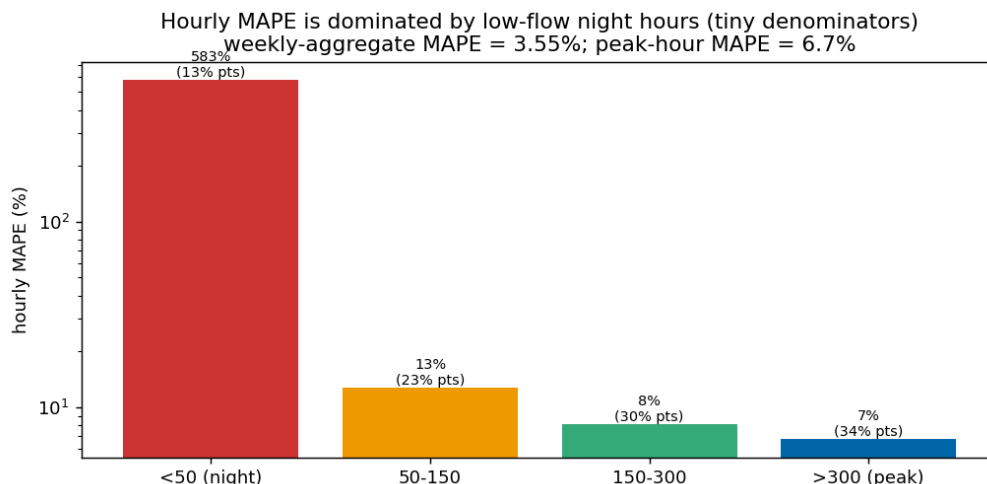


Figure 4: Hourly MAPE by flow regime (log scale). The 82% pooled hourly MAPE is driven by low-flow night hours with tiny denominators; on peak hours it is 6.7%. Weekly aggregation (denominator = weekly total) avoids this entirely.

6.3 Verdict

For Greater Bay Area flow, pre-COVID, the one-week-ahead problem is **effectively solved by a simple, interpretable model**: an EWMA seasonal baseline plus an explicit holiday adjustment, at $R^2 = 0.949$ and weekly-aggregate MAPE 3.55%, sitting at the practical ceiling for endogenous information. The open questions are now about *scope* rather than model class: the small weather lever (Note 03), and generalization beyond this slice – the COVID-era 2020 regime, post-2022 data, other regions, and *speed* (a genuinely harder target at $R^2 \approx 0.76$).

Reproducibility. All numbers and figures come from the `src/` scripts in the project repository (calendar models in `holiday_*` and `ewma_baseline`; §6 headroom in `horizon_ceiling` and `weekly_aggregate_headroom`; §7 feature merit in `feature_merit`; §8 floor and MAPE in `floor_and_mape`).